

IRFAN QAISAR

Control Science and Engineering | Smart Buildings | Occupancy Modeling | HVAC Control

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RESEARCH PROFILE

PhD researcher in Control Science and Engineering at Tsinghua University, specializing in occupancy measurement and prediction, machine learning, and occupant-centric HVAC control for smart buildings. Research combines real-world multi-sensor data with EnergyPlus simulations to evaluate building energy use and thermal comfort. Additional research experience includes robust control, event-triggered control, networked systems, and multi-agent systems. Successfully defended PhD thesis in September 2026.

RESEARCH INTERESTS

Smart Buildings; Occupancy Sensing and Prediction; Occupant-Centric HVAC Control; Machine Learning; Time-Series Modeling; Intelligent Control Building Energy Efficiency; Building Energy Simulation

EDUCATION

Tsinghua University, Beijing, China

PhD Candidate, Control Science and Engineering | Sep 2021 – Sept 2026

Thesis successfully defended, Sep 2026

Supervisor: Prof. Qianchuan Zhao | **Dissertation:** Multi-Sensor Occupancy Sensing for Occupant-Centric Control in Smart Buildings

Nanjing University of Science and Technology, Nanjing, China

M.S., Control Theory and Control Engineering | Sep 2016 - Apr 2019

Dawood University of Engineering and Technology, Karachi, Pakistan

B.E., Electronic Engineering | Jan 2009 - May 2013

RESEARCH EXPERIENCE

PhD Researcher / Research Assistant, Center for Intelligent and Networked Systems (CFINS), Tsinghua University

Sep 2021 – Sept 2026

- Develop machine learning and deep learning models for occupancy inference and time-series prediction using real-world multi-sensor datasets.
- Design Transformer-, Mamba-, and LLM-based frameworks for sequential modeling, classification, regression, and decision support.
- Build control-oriented AI pipelines that connect sensing, prediction, and supervisory HVAC control under simulation and experimental settings.
- Conduct model development, evaluation, and validation using Python, PyTorch, EnergyPlus, OpenStudio, and operational building data. Conducted research across building intelligence and control systems, including robust control, event-triggered control, fuzzy systems, sensor networks, and multi-agent systems.

SELECTED RESEARCH PROJECTS

OCC-Mamba: Indoor Occupancy Prediction

- Developed a Mamba-based model for long-term occupancy trends and short-term dynamics; validated it on Honeycomb, ROBOD, and ASHRAE datasets against MLP, LSTM, Transformer, and Mamba variants.

LLM-Based Occupancy Measurement

- Designed few-shot and structured-prompt methods for occupancy measurement from building sensor data; compared LLMs with conventional models using real-world building datasets from China and Singapore.

Multi-Sensor Occupancy Prediction

- Modeled occupancy presence and occupant counts with Transformers, using occupancy history, indoor environmental measurements, and HVAC operational data in multi-zone office settings.

Occupant-Centric HVAC Control and Energy Evaluation

- Evaluated occupancy-based HVAC strategies in OpenStudio–EnergyPlus; extended the work to surveillance-video occupancy measurement with detection, tracking, and LLM/VLM refinement.

PUBLICATIONS

- **2026**
 1. **Qaisar, I.**, Sun, K., Jia, Q., & Zhao, Q. (2026). Experimental study on surveillance video-based indoor occupancy measurement for occupant-centric control. *Building and Environment*, 115200
 2. **Qaisar, I.**, Sun, K., Zhong, D., Yang, X., Miao, X., & Zhao, Q. (2025). OCC-Mamba: A Mamba-based deep learning approach for indoor occupancy prediction. *Building and Environment*, 114085.
 3. **Qaisar, I.**, Sun, K., & Zhao, Q. (2025). Exploring large language models for indoor occupancy measurement in smart office buildings. *Building and Environment*, 113860.
- **2025**
 4. **Qaisar, I.**, Sun, K., & Zhao, Q. (2025, August). Dynamic Occupancy Measurement for Smart Buildings: A Few-shot Large Language Model Approach. In *2025 IEEE 21st International Conference on Automation Science and Engineering (CASE)* (pp. 2363-2363). IEEE.
 5. **Qaisar, I.**, Liang, W., Sun, K., Xing, T., & Zhao, Q. (2025). An experimental comparative study of energy saving based on occupancy-centric control in smart buildings. *Building and Environment*, 268, 112322.
- **2023**
 6. **Qaisar, I.**, Sun, K., Khan, M. A., Xing, T., & Zhao, Q. (2023). Building occupancy number prediction: A Transformer approach. *Building and Environment*, 244, 110807.
 7. **Qaisar, I.**, Sun, K., Zhao, Q., Xing, T., & Yan, H. (2023). Multi-sensor-based occupancy prediction in a multi-zone office building with transformer. *Buildings*, 13(8), 2002.
 8. Aslam, M. S., & **Qaisar, I.** (2023). Sensor networks with distributed event-triggered scheme for T-S fuzzy system with dissipativity analysis. *European Journal of Control*, 71, 100800.
 9. Shamrooz Aslam, M., **Qaisar, I.**, Majid, A., & Shamrooz, S. (2023). Adaptive event-triggered robust H^∞ control for Takagi–Sugeno fuzzy networked Markov jump systems with time-varying delay. *Asian Journal of Control*, 25(1), 213-228.
 10. Rajput, A. R., Aslam, M. S., Li, Q., **Qaisar, I.**, & Niazi, S. A. (2023). RETRACTED ARTICLE: A defensive design for control application based on networked systems. *Soft Computing*, 27(1), 547-558.
 11. Shamrooz Aslam, M., **Qaisar, I.**, Majid, A., & Shamrooz, S. (2023). Adaptive event-triggered robust H^∞ control for Takagi–Sugeno fuzzy networked Markov jump systems with time-varying delay. *Asian Journal of Control*, 25(1), 213-228.
- **2022**
 12. **Qaisar, I.**, & Zhao, Q. (2022). Energy baseline prediction for buildings: A review. *Results in Control and Optimization*, 7, 100129.
 13. PANDEY, H. M., Aslam, M. S., Tiwari, P., Band, S. S., **Qaisar, I.**, & El Sayed, H. (2022). Extended Sliding Mode Observer for Delayed TS Fuzzy Control with Non-PDC Approach. *IEEE Transactions on Fuzzy Systems*, TFS-2021.
- **2021**
 14. Aslam, M. S., **Qaisar, I.**, Majid, A., & Ramaraj, P. (2021). Design of sliding mode controller for sensor/actuator fault with unknown input observer for satellite control system. *Soft Computing*, 25(24), 14993-15003.
 15. Xia, C., **Qaisar, I.**, Aslam, M. S., & Qiaoyu, L. (2021, September). Tracking Control of Neural System using Adaptive Sliding Mode Control for Unknown Nonlinear Function. In *2021 IEEE 4th International Conference on Information Systems and Computer Aided Education (ICISCAE)* (pp. 157-161). IEEE.
 16. Aslam, M. S., **Qaisar, I.**, Majid, A., & Ramaraj, P. (2021). Design of sliding mode controller for sensor/actuator fault with unknown input observer for satellite control system. *Soft Computing*, 25(24), 14993-15003.
- **2020**
 17. Chunrong, X., **Qaisar, I.**, Aslam, M. S., & Qiaoyu, L. (2020, November). Reliable filtering for Markovian Jump System Under the Networked Control System. In *2020 Chinese Automation Congress (CAC)* (pp. 7559-7565). IEEE.
 18. **Qaisar, I.**, Aslam, M. S., & Zhou, C. (2020). Event-triggered based H_∞ Consensus Control for Multi-agent Systems under Time-varying Delay. *Journal of Control Engineering and Applied Informatics*, 22(3), 25-32.
 19. Aslam, M. S., **Qaisar, I.**, & Saleem, M. A. (2020). Quantized event-triggered feedback control under fuzzy system with time-varying delay and actuator fault. *Nonlinear Analysis: Hybrid Systems*, 35, 100823.
- **2019**
 20. Ahmed, S., Wang, H., Aslam, M. S., Ghous, I., & **Qaisar, I.** (2019). Robust adaptive control of robotic manipulator with input time-varying delay. *International Journal of Control, Automation and Systems*, 17(9), 2193-2202.

21. **Qaisar, I.**, Zhou, C., Shi, Z., Xu, W., & Xu, P. **(2019, June)** . Exponential Stability for Event-triggered Consensus Control of Heterogeneous Multi-Agent Systems. In *2019 Chinese Control And Decision Conference (CCDC)* (pp. 2373-2378). IEEE.

PATENT

Waste Heat Recovery Optimization Control System and Method, CN 2024105441305, filed Apr 30, 2024. Co-inventor with Tsinghua University and Hitachi Ltd.

TECHNICAL SKILLS

Programming & scientific computing: Python, MATLAB, Pandas, NumPy, Matplotlib, LaTeX.

Machine learning: PyTorch, deep learning, time-series modeling, Transformer, Mamba, LLMs/VLMs, sensor fusion.

Building simulation: EnergyPlus, OpenStudio, Modelica; energy and thermal-comfort evaluation.

Control & automation: Occupant-centric HVAC control, robust and event-triggered control, networked control, multi-agent systems, fuzzy systems, PLC programming, HMI systems.

ACADEMIC SERVICE

Peer Reviewer: *Applied Energy; Automation in Construction; Building and Environment; Energy and Buildings; Engineering Applications of Artificial Intelligence; Expert Systems with Applications; ISA Transactions; Journal of Building Engineering; MethodsX; Results in Control and Optimization.*

HONORS & SCHOLARSHIPS

- Belt and Road Ambassador Scholarship, Overseas Chinese Charity Foundation of China, 2026
- Tsinghua University Comprehensive Excellence Scholarship (Second Class), 2026, 2025 and 2024
- Chinese Government Scholarship, Tsinghua University, 2021 - Present
- Tsinghua-Nike Sustainability Ambassador, 2024
- Logic Star Award, ESG Innovation Case Analysis Roadshow, 2024
- Nanjing Municipal Government Scholarship, 2016 - 2019

EARLIER INDUSTRY EXPERIENCE

Electrical Engineer | Wuxi Johnnywell Railway Equipment Technology Co., Ltd., China

Jun 2019 – Jul 2021. PLC programming, HMI systems, automation troubleshooting, and industrial workflow improvement.

Management Trainee | Design Line Architects, Pakistan | Jun 2014 – Jun 2016

Trainee Engineer | Niagara Textile (Pvt) Ltd., Pakistan | May 2013 – May 2014